YOUSEF ALI AICHA

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SUMMARY

I design embedded systems from custom PCB through firmware to the software that consumes the resulting data, with hands -on experience taking ESP32 and ATmega328P systems from schematic to deployment.

EDUCATION

IE UNIVERSITY **Madrid, Spain**
Bachelor of Science in Computer Science and Artificial Intelligence **Expected: July 2029**

- GPA 8.53/10 cumulative.
- Recipient of the 100% tuition UWC Scholarship and the Horizon Foundation Living Grant.

UNITED WORLD COLLEGE IN MOSTAR (UWCiM) **Mostar, Bosnia and Herzegovina**
International Baccalaureate (IB) — Score: 36/45 **Graduated: May 2025**

SKILLS

- **Software**: C, C++, JavaScript, Git.
- **Embedded**: ESP32-S3, ATmega328P, Arduino, KiCad.
- **ML/Data**: Python, LightGBM, NumPy, Streamlit.
- **Languages**: Arabic (Native), English (Fluent), Turkish (Fluent), Spanish (A2).
- **Tools**: Git

TECHNICAL PROJECTS

AEGIS — Networked Rotating Ultrasonic Radar System AEGIS

- Built a radar on an ESP32-S3: an HC-SR04 performs ultrasonic ranging to 300cm while a PWM-driven servo sweeps it in 2° increments, broadcasting telemetry over WebSocket at ~40Hz to a browser dashboard served entirely from the board's own flash.
- Designed a custom KiCad PCB and a non-blocking firmware architecture with runtime-adjustable sweep parameters and click-to-steer control. Full 180° sweep cycle in ~9s; 68% program storage utilization.

SENTINEL — Adaptive Traffic-Light Finite State Machine SENTINEL

- Implemented a four-state traffic controller on bare ATmega328P with zero external libraries: polling loop gated entirely by millis() comparisons, no delay() calls, full 65s signal cycle, 2ms-multiplexed 7-segment display via 74HC595. 36% program storage utilization.
- Added a light-sensing night mode (ADC-sampled) that detects an idling vehicle and adaptively shortens the following phases, with UART serial diagnostics at 9600 baud. Backed by a custom KiCad PCB.

EXPERIENCE

Teknofest Electric Vehicle Initiative Gaziantep, Turkey

Founder and Project Lead — August to December 2021

- Led a 12-person student team through the design, budgeting, and prototyping of a full-scale electric vehicle chassis in CAD.
- Managed external sponsorships, placing within five points of the qualifying threshold for financial aid.

Institute of the Middle East Aleppo, Syria

Project Communications and Data Officer — June to September 2025

- Coordinated communication pipelines and translated project documentation across Arabic, English, and Turkish for distributed regional development teams.
- Developed a Datathon network optimization strategy for electric vehicle charging station deployment across Spain.

LEADERSHIP & ACTIVITIES

- DENEYAP Technology Scholar — completed 11 certified courses in design, robotics, electronics, and aeronautics as part of Turkey's first-generation elite cohort; State Honors for Best Eco-Friendly Robot.
- Founder, CS and Engineering Club (UWC Mostar) — restructured the club around student-led coding sessions and built its project

showcase site.